

p1.py
model create

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from pickle import dump

data = pd.read_csv("cs1.csv")
print(data)

feature = data[["area in square feet"]]
target = data["price (in crores)"]

model = LinearRegression()
model.fit(feature.values, target)

with open("hpp.pkl", "wb") as f:
    dump(model, f)
    print("model saved")
```

p2.py
model use

```
from pickle import load

with open("hpp.pkl", "rb") as f:
    model = load(f)
    print("model ready")

area = float(input("enter area "))
price = model.predict([[area]])
msg = "₹ " + str(round(price[0], 2)) + " crores "
print(msg)
```


Create web app:

Step 1: Project and Files

```
C:\mljuly26\L18>mkdir cs1_hpp_app
C:\mljuly26\L18>cd cs1_hpp_app
C:\mljuly26\L18\cs1_hpp_app>notepad app.py
C:\mljuly26\L18\cs1_hpp_app>mkdir templates
C:\mljuly26\L18\cs1_hpp_app>cd templates
C:\mljuly26\L18\cs1_hpp_app\templates>notepad home.html
```

Step 2: Coding

app.py

```
from flask import *
from pickle import load
import os
fn = "hpp.pkl"

if os.path.exists(fn):
    with open(fn, "rb") as f:
        model = load(f)
else:
    print(fn, " not found")

app = Flask(__name__)

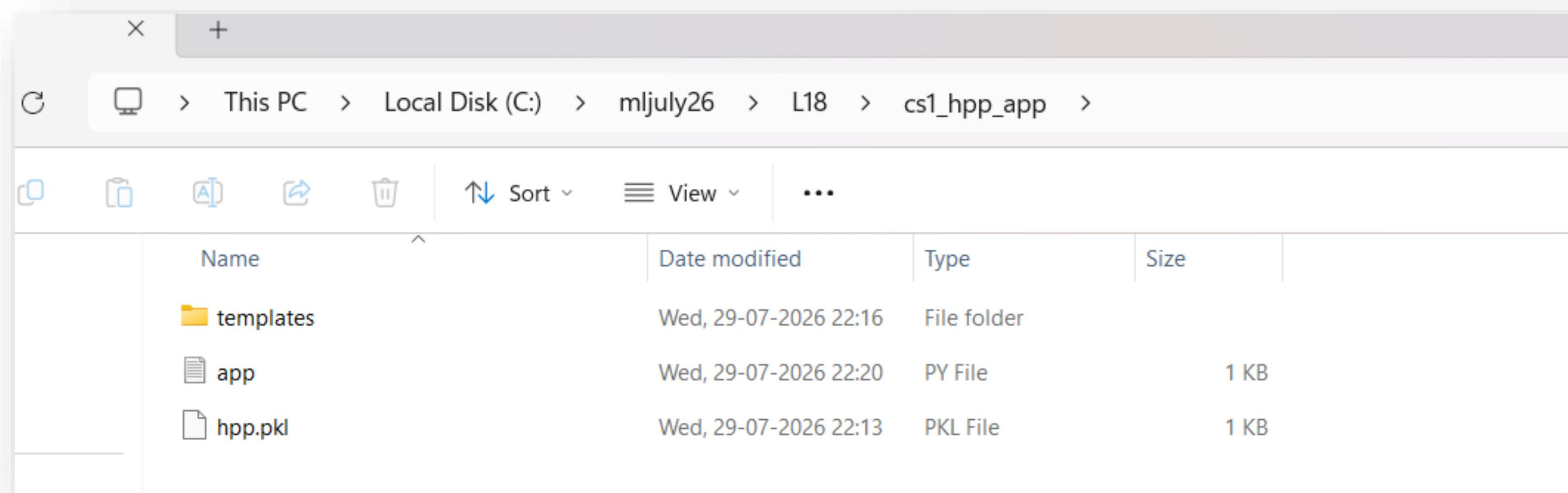
@app.route("/", methods=["POST", "GET"])
def home():
    if request.method == "POST":
        area = float(request.form.get("area"))
        price = model.predict([[area]])
        msg = "₹ " + str(round(price[0], 2)) + " crores "
        return render_template("home.html", msg=msg)
    else:
        return render_template("home.html")

if __name__ == "__main__":
    app.run(debug=True, use_reloader=True)
```


home.html

```
<html>
  <head>
    <title>
      Lonavala Housing Price Predictor App by Kamal Sir
    </title>
    <style>
      * {
        font-size: 30px;
        text-align:center;
        font-family: Cambria;
      }
      body {
        background-color: linen;
      }
      .fc {
        border:solid;
        padding:2%;
        width:50%;
        margin:auto;
        border-radius:30px;
        background-color:white;
      }
    </style>
  </head>
  <body>
    <h1> Lonavala Housing Price Predictor </h1>
    <div class="fc">
      <form method="POST">
        <label> Area </label>
        <input type="number"
          step="0.01"
          name="area"
          placeholder="Enter Area in Square Feet"
          min="1000"
          required />
        <br/><br/>
        <input type="submit"
          value="Predict Price"/>
      </form>
    </div>
    <h2>
      {{ msg }}
    </h2>
  </body>
</html>
```


Step 3: mode the.pkl file into web application folder



Step 4: run the app

```
C:\mljuly26\L18\cs1_hpp_app\templates>cd..  
C:\mljuly26\L18\cs1_hpp_app>python app.py
```

Step 5: deploy